



Day No	Session No	Learning Time (Hours)	Module Name	Topic	Sub-Topic	Remarks
Day 01	Session 01	0.5	AWS Cloud Overview	Project Briefing	Quick Overview : Agenda & Objective : Project Overview & Activities : a. Present Business scenario - Deploying a Full Stack Application manually leveraging the AWS Service - EC2, S3, RDS (e.g., a simple real-time problem) b. Functional Requirements break down c. Understand the business problem and outline the prototype's features. d. Discussion on Technical Pre-requisites e. Walkthrough of Cloud Models and architecture	Faculty to give an overview and explain the important aspects of these topics from implementation perspective through hands-on
		0.3		AWS Cloud Overview	Quick Overview : Explain a. Basics of cloud computing. b. IaaS, PaaS, and SaaS models. c. How AWS provides these services? d. Comparison of Cloud with Traditional on-premises IT. e. Get an overview of AWS and its global infrastructure. f. Explore major AWS services and their use cases. g. Navigating the AWS Management Console. Project Task : Explore the understanding of AWS cloud fundamentals by completing the following activities Activities: a. List examples of IaaS, PaaS, and SaaS in AWS. b. Discuss pros and cons of cloud vs. on-premises. c. Map a sample business requirement to the right cloud model. d. Identify AWS regions and services available in each region. e. Log in to the AWS Console and locate key services. Match business scenarios to appropriate AWS services.	
		0.3				
		0.3	AWS Identity and Access Management (IAM)	AWS Identity and Access Management (IAM)	Quick Overview : Explain how AWS Identity and Access Management (IAM) is an AWS service enabling secure management of access to AWS resources like users, groups, roles, and defines granular permissions using policies, apart from best practices.	
		0.3			Project Task: Understand IAM Service features Activities: The following activities are just from a theoretical perspective, making trainees understand how IAM works, but they will not be able to do anything in sandboxes due to restricted access. a. Understanding IAM Users and Groups b. Organize users into groups based on job functions (e.g., admins, developers). c. Attach Policies to Users and Groups d. Create and Assign IAM Roles e. Define roles for specific tasks or services (e.g., EC2 instance role). f. Apply the Principle of Least Privilege	
		0.25	AWS Regions and Availability Zones	AWS Regions and Availability Zones	Quick Overview : a. Learn how AWS organizes its data centers using regions and AZs. b. Understand how to select regions for deployment. c. Discuss high availability and disaster recovery strategies. Project Task: Implement Parent-Child Components	
		0.25			Activities: a. Choose a region for a sample deployment. b. Design a simple high-availability architecture using multiple AZs. c. Simulate a failover scenario and discuss outcomes.	
		0.25	AWS Regions and Networks	Virtual Private Cloud (VPC) and Networking	Quick Overview : a. Introduction to VPC and its core components. b. Learn about subnets, route tables, gateways, and CIDR blocks. c. Understand security groups and network ACLs. Project task : Explore the activities related to the VPC mentioned	
		0.25			Activities : These activities again depend on the availability of access to the sandbox. If not, these concepts will be learnt from understanding perspective only a. Create a VPC with public and private subnets. b. Set up route tables and an internet gateway. c. Configure security groups for a web server and a database. d. Experiment with network ACLs to restrict traffic.	
		0.25	EC2 Service	Launching an EC2 Instance	Quick Walkthrough : Why Amazon EC2, benefits, the process and steps involved in leveraging EC2 Service in AWS for launching, configuring, and managing virtual servers for specific computing needs Project Task : Exploring the process to create a DB using RDS and configure it as per the Business need Activities : a. Sign in to the AWS Management Console. b. Navigate to the EC2 Dashboard. c. Click "Launch Instance." d. Choose an Amazon Machine Image (AMI). e. Select an appropriate instance type for a given workload. f. Configure instance details (number of instances, network, etc.). g. Add storage (EBS volumes). h. Add tags (optional). i. Configure security group (set inbound/outbound rules). j. Review and launch the instance. k. Select or create a key pair for SSH access. l. Launch the instance and wait for its status to become "running." m. Further explore instance types and pricing. n. Configure security groups to allow HTTP, SSH traffic, and key pairs for access.	
		0.3			Quick Walkthrough : Provide a quick understanding of the importance of connecting to the EC2 instance and the steps involved to do so Project Task : Exploring the different ways to connect to an EC2 Instance	
		0.3		Connecting to the EC2	Activities : a. SSH (Linux/Unix Instances) Use an SSH client with the private key file (.pem) to connect from a terminal. Example command: <code>ssh -i "your-key.pem" ec2-user@public-dns-name</code> b. EC2 Instance Connect: Use the AWS Management Console's "Connect" button for browser-based SSH access. c. Session Manager (via AWS Systems Manager) Connect without SSH keys using IAM permissions and the AWS Console or CLI. d. Third-Party Tools Use tools like PuTTY (after converting the key to .ppk format) for SSH access.	
		0.3				

Session 02	0.3	RDS		Quick Overview: Why Amazon RDS, benefits, the process and steps involved in leveraging RDS Service in AWS for creating a specific DB Instance Project Task : Exploring the process to create a DB using RDS and configure it as per the Business need Activities : - Log in to the AWS Management Console. - Navigating to the RDS Service Dashboard using Amazon Web Console - Click "Create Database." - Select "Standard Create" as the creation method. - Choose the type of DB Instance (e.g., "MySQL") that has to be created as the database engine. - Pick the desired version of the DB Instance (e.g., MySQL). - Select a template (e.g., Free tier). - Enter a DB instance identifier, master username, and password. - Choose the DB instance class (e.g., db.t3.micro). - Set allocated storage and autoscaling options (if needed). - Configure VPC, public accessibility, and security groups. - Set additional configurations like database name and backups(as per the need). - Review your settings and click "Create Database." - Wait for the instance status to become "Available." - Leverage the the endpoint and connect using a MySQL client to verify the created DB Instance in Cloud	
	0.5				
	0.3	RDS Service	Deploying Spring Boot on EC2 with RDS integrated	Quick Walkthrough : Understand the process and steps involved in deploying a SpringBoot App onto and EC2 while integrating it with RDS already created Project Task : Activities : - Connect to EC2 Using SSH : - Access the EC2 instance securely from your local machine using SSH and the appropriate key pair. - Install Java and Set Up the Environment: - Update the package manager and install Java (e.g., OpenJDK). - Verify the Java installation and configure environment variables if needed. - Upload and Configure a Spring Boot Application: - Transfer the Spring Boot.JAR file to the EC2 instance using SCP or SFTP. - Set necessary permissions for the application file. - Set Environment Variables for RDS Connectivity - Export environment variables or update the Spring Boot APP configuration to include AWS RDS endpoint, database name, username, and password. - Run the SB Application and Connect It to RDS: - Start the Spring Boot application using the Java command. - Ensure the application is configured to connect to the RDS database using the provided credentials and endpoint. - Test the Application's Connection to RDS - Access the application (via browser or API endpoint) and verify it can read/write data to the RDS database. - Troubleshoot Common Deployment Issues - Diagnose and resolve issues such as security group misconfigurations, incorrect database credentials, or application errors. - Check EC2 and RDS logs for error messages. - Adjust security group rules to allow traffic between EC2 and RDS if needed.	
	0.3				
	0.67	S3 and Static Website Hosting	S3 and Static Website Hosting	Quick Overview : Discuss the purpose, benefits, features of S3 Service in AWS, and how it can be leveraged for Static Website Hosting Project Task : Activities : - Create an S3 bucket with versioning enabled. - Set a bucket policy to allow public read or restrict access. - Generate a pre-signed URL for file upload/download. - Configure API Endpoints in UI Application for Backend Integration. In your UI APP environment configuration, set the REST API base URL to point to your deployed Spring Boot backend (hosted on EC2). - Build the UI - Angular/React Application on local machine - Upload Build Artifacts to the S3 Bucket - Access the Deployed Static Website with the S3 static website endpoint URL in your browser to view the deployed application.	
	0.3	Full Stack Application in Cloud	Full Stack App in Cloud - Testing	Quick Overview : Discuss the workflow involved in deploying the front-end, back-end, database, and every component of Full Stack application, which talk to each other, in the Cloud. Perform the end-to-end testing of the FSD app in the Cloud. Project Task : Test the Full Stack Application with every component deployed in cloud Activities : - Verify Frontend Accessibility - Open the frontend application's public URL (e.g., S3 static website endpoint). - Ensure the UI loads without errors and static assets are served correctly. - Test User Authentication and Authorization - Attempt to log in or register using valid and invalid credentials. - Confirm authentication flows work and unauthorized access is blocked where appropriate. - Check API Connectivity from Frontend - Trigger frontend actions that make API calls (e.g., form submissions, data fetches). - Use browser developer tools to inspect network requests and ensure correct API endpoints are called. - Validate Data Flow and CRUD Operations - - Create, read, update, and delete (CRUD) data through the UI. - Confirm changes are reflected and persisted in the database. - Cross-Component Data Consistency - After performing operations (e.g., adding a record), refresh the UI or access data from another client to ensure consistency. - Query the database directly (if possible) to verify data integrity.	
	0.5				
	0.5	Containerization Overview in AWS	Project Briefing	Quick Overview : Agenda & Objective : Project Overview & Activities : - Present Business scenario - Deploying a Full Stack application leveraging the Containerization Workflow with Services like ECR, ECS while understanding Dockerizing the applications (e.g., a simple real-time problem) - Functional Requirements break down - Understand the business problem and outline the prototype's features. - Discussion on Technical Pre-requisites - Walkthrough of Containerization	
	0.5			Quick Overview : - Learn what containerization is and how it evolved. - Understand the differences between containers and virtual machines. - Explore the benefits: portability, consistency, efficiency, and scalability. - Compare container and VM architectures. - Identify key container components and lifecycle stages.	

Day 02	Session 01	0.5		Introduction to Containerization	Project Task : Understand and perform the activities related to Containerization Activities : a. List three ways containers differ from VMs. b. Research and present a real-world use case for containers. c. Draw a diagram comparing container and VM architectures. d. Identify and label the main components in a container ecosystem (image, container, registry, orchestrator).	Faculty to give an overview and explain the important aspects of these topics from implementation perspective through hands-on
		1		Docker Fundamentals	Quick Overview : a. Understand Docker's architecture: Engine, Daemon, CLI, and Hub. b. Learn what Docker images and containers are. c. Explore how Docker uses a layered file system. d. Practice essential Docker commands for managing containers. e. Learn to build images with Dockerfile and commit changes.	
		1			Project Task : Activities : a. Install Docker and run the hello-world container. b. List all Docker images and running containers. c. Build a simple Docker image using a Dockerfile. d. Use Docker commands to start, stop, and remove containers. e. Commit changes from a running container to a new image.	
		0.5		Containerizing Spring Boot Applications	Quick Overview : Prepare a Spring Boot app for containerization by reviewing dependencies. Write a Dockerfile step-by-step for the application. Build Docker images and tag them. Run containers with port mapping. Enable communication between multiple containers using Docker networks.	
	Session 02	1			Project Task : Dockerize the SB App Activities : a. Create a Dockerfile for a sample Spring Boot app. b. Build and tag the Docker image. c. Run the container and access the app via browser. d. Set up a Docker network and connect two containers. e. Demonstrate service discovery between containers.	
		0.5		Introduction to CloudWatch and Monitoring	Quick Overview : a. Learn about CloudWatch metrics for AWS resources. b. Organize logs using log groups and streams. c. Send application logs to CloudWatch. d. Set up alarms for resource thresholds and integrate notifications.	
		1			Project Task : Implement and test the CloudWatch Activities: a. Monitor EC2 or container metrics in CloudWatch. b. Create a log group and stream application logs. c. Set up a CloudWatch alarm for high CPU usage.	
		0.3		Introduction to Lambda and Event-Driven Integration	Quick Overview: Discuss AWS Lambda - its usage, features, and benefits	
		1.17			Project Task: Implement and Test the lambda Function Activities : a. Deploy your Spring Boot application and ensure its API is accessible via an HTTP endpoint. b. Create an AWS Lambda function using your preferred runtime (e.g., Java, Python, Node.js). c. Configure the Lambda function with necessary permissions to access external APIs. d. Set up an event source (such as API Gateway, S3, or DynamoDB) to trigger the Lambda function. e. Instruct the Lambda function to make HTTP requests to the Spring Boot API endpoint when triggered. f. Handle the API's response and errors within the Lambda function's logic.	
	Session 01	0.5		Introduction to ECR and ECS	Quick Overview : a. Learn what Amazon ECR is and its role as a managed Docker registry. b. Understand why ECR is used for secure image storage and sharing. c. Explore Amazon ECS as a managed container orchestration service. d. Compare ECS with Kubernetes and Docker Swarm. e. See how ECR and ECS work together for container deployments.	Faculty to give an overview and explain the
		1			Project Task : Activities : a. List the benefits of using ECR over self-hosted registries. b. Compare ECS and Kubernetes in a table. c. Diagram explaining the flow of a Docker image from ECR to ECS. d. Identify scenarios where ECS is a better fit than other orchestrators.	
		0.5		Deploying Spring Boot Microservices on ECS	Quick Overview : a. Prepare Spring Boot microservices by packaging them as Docker images. b. Push these images to ECR repositories. c. Create an ECS cluster to run your containers. d. Define ECS task definitions to specify how containers run. e. Deploy services and configure auto-scaling. f. Set up networking and load balancing with an Application Load Balancer. g. Monitor and troubleshoot deployments using CloudWatch.	
		2			Project Task : Activities : a. Dockerize a sample Spring Boot microservice. b. Push the Docker image to an ECR repository. c. Create an ECS cluster and register a task definition. d. Deploy the service and set up auto-scaling policies. e. Attach the service to an Application Load Balancer. f. View logs and metrics for the service in CloudWatch. g. Simulate a scaling event and observe ECS behavior.	
					(Optional based on the time saved as the training progress from Day-1)	
		0.5			Quick Overview : Introduction to DevOps, Explain how CI/CD can be implemented leveraging the AWS Service - CodeBuild and Code Pipeline that automates the Continuous Building, Continuous Integration, and Continuous Deployment	

Day 03	Session 02	2.5	DevOps and CI/CD	CI/CD - CodeBuild, CodePipeline	(Optional based on the time saved as the training progress from Day-1) Project Task : Set up an automated CI/CD pipeline using AWS CodePipeline and CodeBuild to build, test, and deploy a sample application. Activities : - Create a Source Repository - Set up a source code repository (e.g., GitHub). - Configure AWS CodeBuild Project - Define a build specification file (buildspec.yml). - Set up CodeBuild to pull code from the repository and run build/test commands. - Set Up AWS CodePipeline - Create a new pipeline and add stages for source, build (using CodeBuild), and deploy. - Integrate CodeBuild as the build provider within the pipeline. - Automate Deployment - Add a deployment stage (e.g., AWS Elastic Beanstalk, ECS, S3) to the pipeline. - Trigger Pipeline on Code Change - Configure the pipeline to automatically start on code commits or pull requests. - Monitor and Debug Pipeline Runs - Use AWS Console to monitor pipeline execution and review build/test logs for troubleshooting. - Test End-to-End Workflow - Make a code change, push it to the repository, and verify that the pipeline builds, tests, and deploys the application automatically.	Faculty to give an overview and explain the important aspects of these topics from implementation perspective through hands-on
		0.5	Deployed Microservices Solution Review	End-to-End Solution Review	Quick Overview : - Review the overall architecture of the deployed microservices solution. - Walk through each deployment step, from building images to ECS deployment. - Discuss best practices for security, scalability, and monitoring. - Identify common deployment challenges and solutions. - Engage in Q&A and discuss next steps for further learning. Project Task : Implement Route Params, Redirecting, and PageNotFound scenarios	
		0.5			Activities : - Draw the architecture diagram for the complete solution. - Present a step-by-step recap of the deployment process. - List best practices applied during the project. - Identify and troubleshoot a simulated deployment issue. - Prepare questions for group discussion and suggest improvements or next steps.	
Day 04	Session 01	4	Assignment	Assignment + Track-End Quiz Assessment		-
	Session 02	4	Assignment	Assignment + Track-End Quiz Assessment		-
Day 05	Session 01	4	Assignment	Assignment + Track-End Quiz Assessment		-
	Session 02	4	Assignment	Assignment + Track-End Quiz Assessment		-